Decentralized Multi-Authority FE
with Malicious Authorities from Standard Assumptions

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Multi-Authority Functional Encryption

[Chase07, Chase-Chow09, Lewko-Waters11, Brakerski-Chandran-Goyal-Jain-Sahai-Segev17]







MSK_1 MSK_2 MSK_n

$SK_{GID,x[1]} = KGen(MSK_1, GID, x[1])$

$SK_{GID,x[2]} = KGen(MSK_2, GID, x[2])$

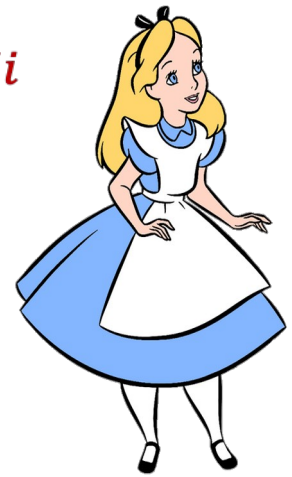
$SK_{GID,x[n]}$

$\{MPK_i\}_i$

f

$CT = Enc(\{MPK_i\}_i, f)$

GID





Wants to learn $f(x)$

*$x[i]$ is the ith bit of x

Role of GIDs

[Chase07, …]

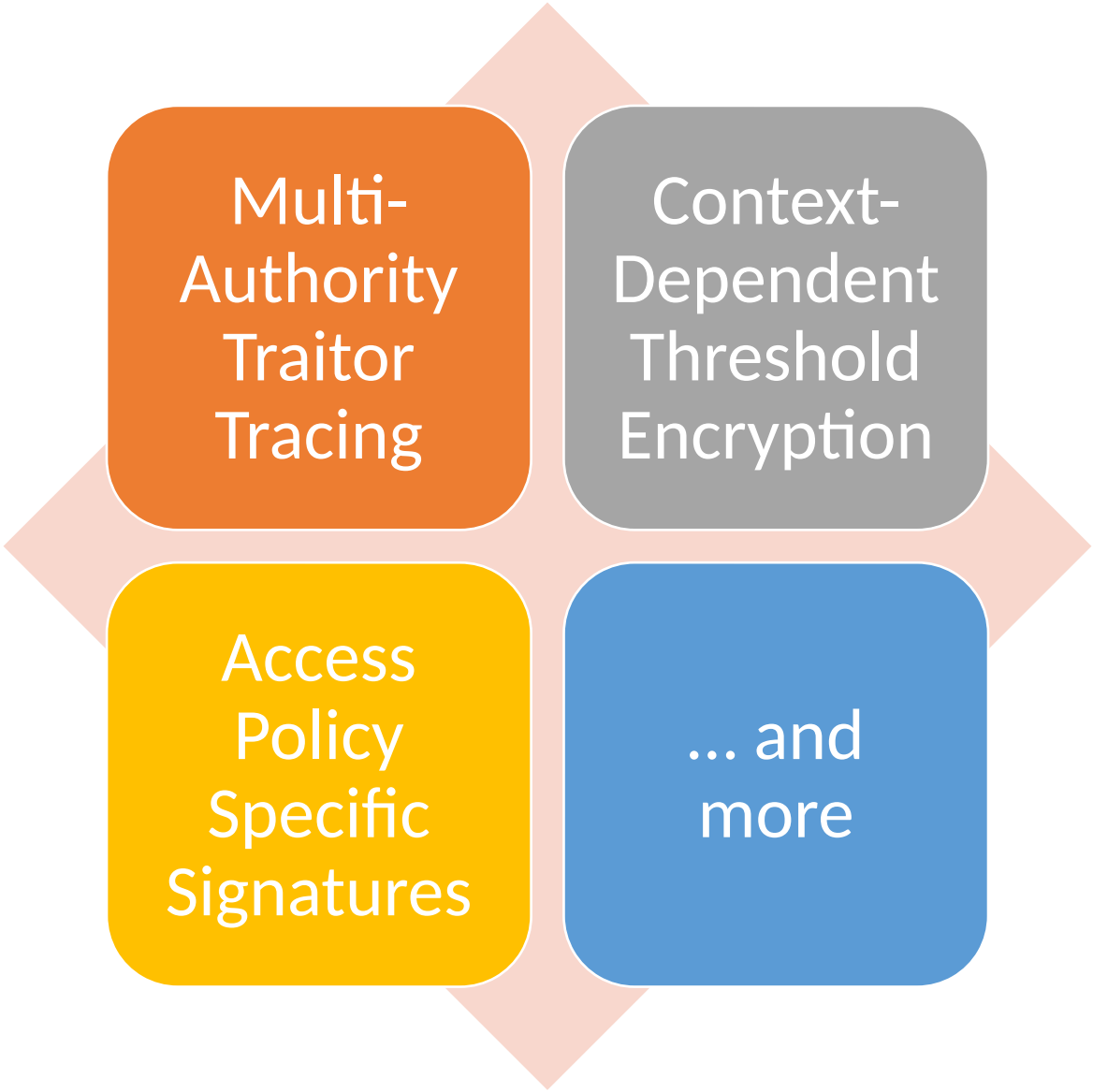
- Different receivers have different inputs
  - Charlie gets key for x and Dave gets for y
  - i.e., Charlie gets $\{SK_{x[i]}\}_i$ and Dave gets $\{SK_{y[i]}\}_i$
  - Should not be possible to "*mix*" their keys.

  - Classic approach: annotate each key with unique global identifier, GID
  - E.g., GID is SSN/passport

Other Model for Decentralized FE

Registration-Based Encryption	[Garg-Hajiabadi-Mahmoody-Rahimi'18, Garg-Hajiabadi-Mahmoody-Rahimi-Sekar'18, Goyal-Vusirikala'20 …]
Others	[Chen-Harrison-Soldera-Smart'02, Goyal'07, …]

Theoretical Use Cases



Multi-Authority: Security

[Chase07, …, Brakerski-Chandran-Goyal-Jain-Sahai-Segev17]

INTUITION

Given $Enc(\{MPK_i\}_i, f)$ and $\{SK_{GID_1,x_1[i]}\}_i, \ldots, \{SK_{GID_q,x_q[i]}\}_i$

Adversary learns **only** $f(x_1), \ldots, f(x_q)$

In addition, Adversary can also receive $\{MSK_j\}_j$

Bounded Collusion Model

Secure as long as < Q collusions

Landscape

- For General Circuits
  - Full collusion resistant
  - From sub-exponential $i\mathcal{O}$ + sub-exponential OWF [Brakerski-Chandran-Goyal-Jain-Sahai-Segev17]

  - Bounded collusion <u>without Authority Corruptions</u>
  - From sub-exponential PKE + ROM [Goyal-Y24]

<u>Open Question</u>

Can we construct MAFE from polynomially hard assumptions?

Our Results



MAFE for General Circuits*

- Assumption: LWE or SXDH
- Bounded Collusions

Prior work: None

Generic Compiler*

Can be extended to

Collusion Resistant MAFE + FE

MAFE for General Circuits

- Without Authority Corruptions
- Assumption: PKE
- Bounded Collusions

Prior work: Sub-exp PKE + ROM [Goyal-Y'24]

New Lower Bounds

* For certain parameter ranges

In this talk…

NO RANDOM ORACLES

MAFE for General Circuits*

- Assumption: LWE or SXDH
- Bounded Collusions

Prior work: None

Generic Compiler*

Can be extended to, Collusion Resistant MAFE + FE

MAFE for General Circuits

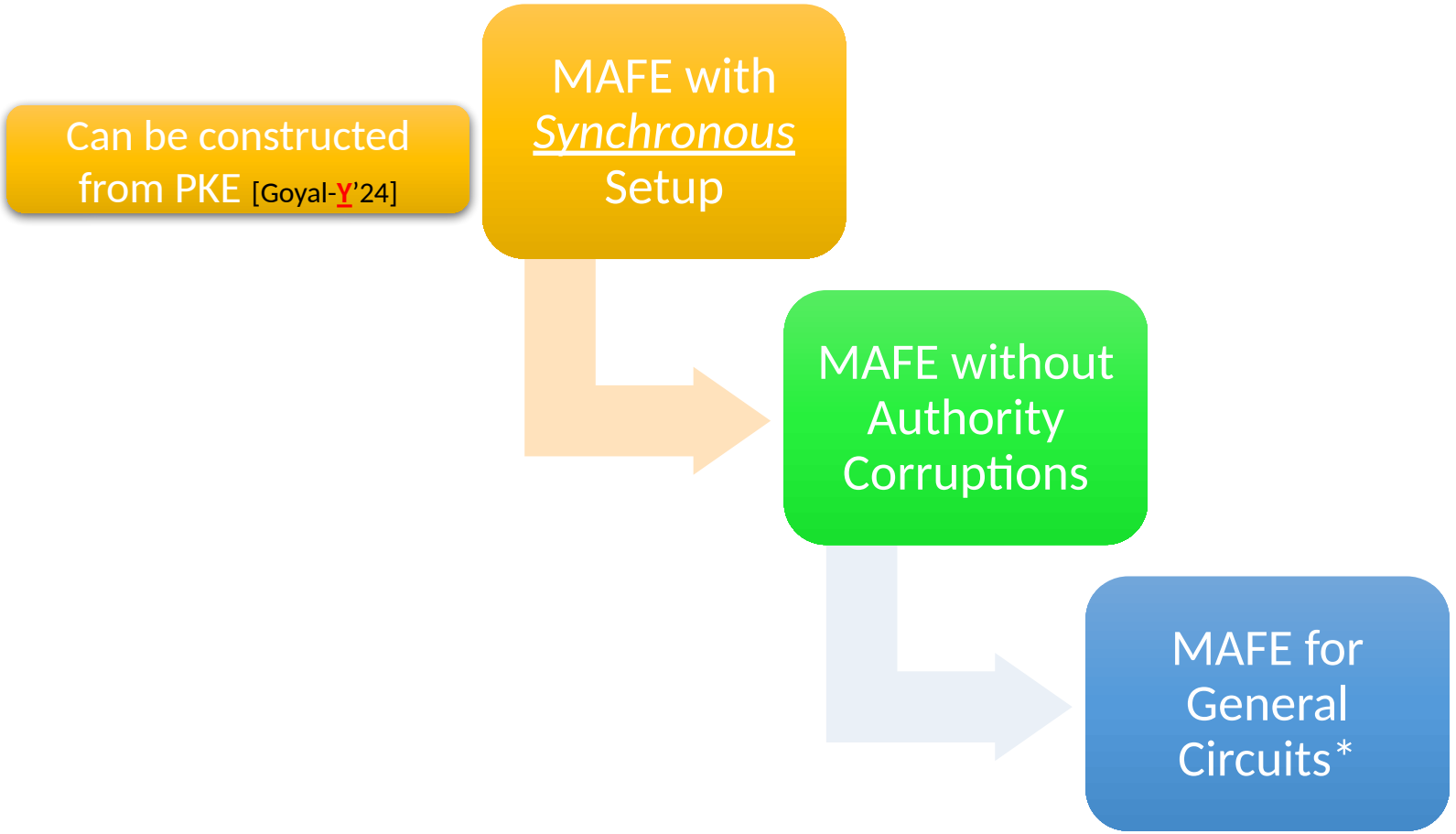
- Without Authority Corruptions
- Assumption: PKE
- Bounded Collusions

Prior work: Sub-exp PKE + ROM [Goyal-Y'24]

New Lower Bounds

* For certain parameter ranges

Road Map



* For certain parameter ranges

MAFE (without Authority Corruptions)







MSK_1

MSK_2

MSK_n

$SK_{GID,x[1]} = KGen(MSK_1, GID, x[1])$

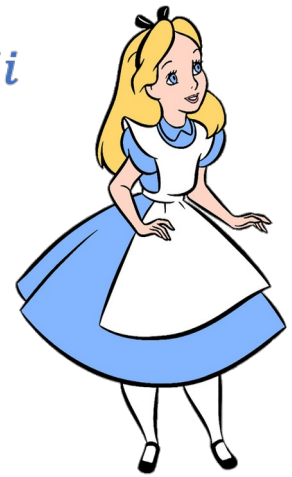
$SK_{GID,x[2]}$

$SK_{GID,x[n]}$

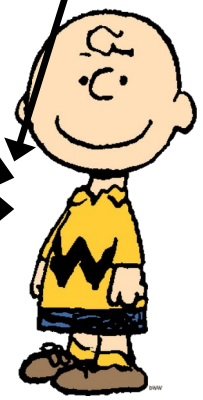
GID

$\{MPK_i\}_i$

f



$CT = Enc(\{MPK_i\}_i, f)$



MAFE with *Synchronous* Setup



Global Setup







MSK_1 K MSK_2 K MSK_n K

$SKGID, x[1] = KGen(MSK_1, GID, x[1])$

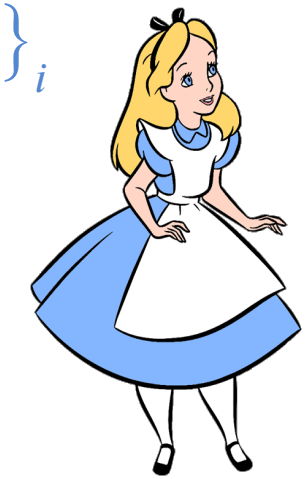
$SKGID, x[2]$

$SKGID, x[n]$

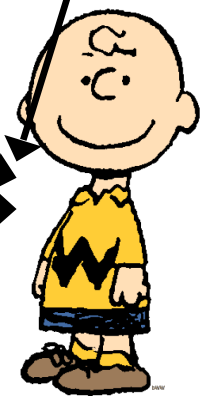
GID

$\{MPK_i\}_i$

f



$CT = Enc(\{MPK_i\}_i, f)$



Can be constructed from PKE [Goyal-Y'24]

From MAFE with Synchronous Setup to MAFE

Each Authority simply samples 1-Auth FE

From PKE
[Gorbunov-Vaikunatanathan-Wee12],[Ananth-Vaikuntanathan19]

$\text{AuthSetup}(i) \rightarrow$ Run 1-Auth FE Setup

MPK_i MSK_i

$\text{KGen}(i, GID, x[i]) \rightarrow$ Run 1-Auth FE Key Generation



Encryption Does the Hard Work

$\text{Enc}(MPK_1 \ldots MPK_n, f) \rightarrow$

- Sample MAFE with synchronous setup Instantiation

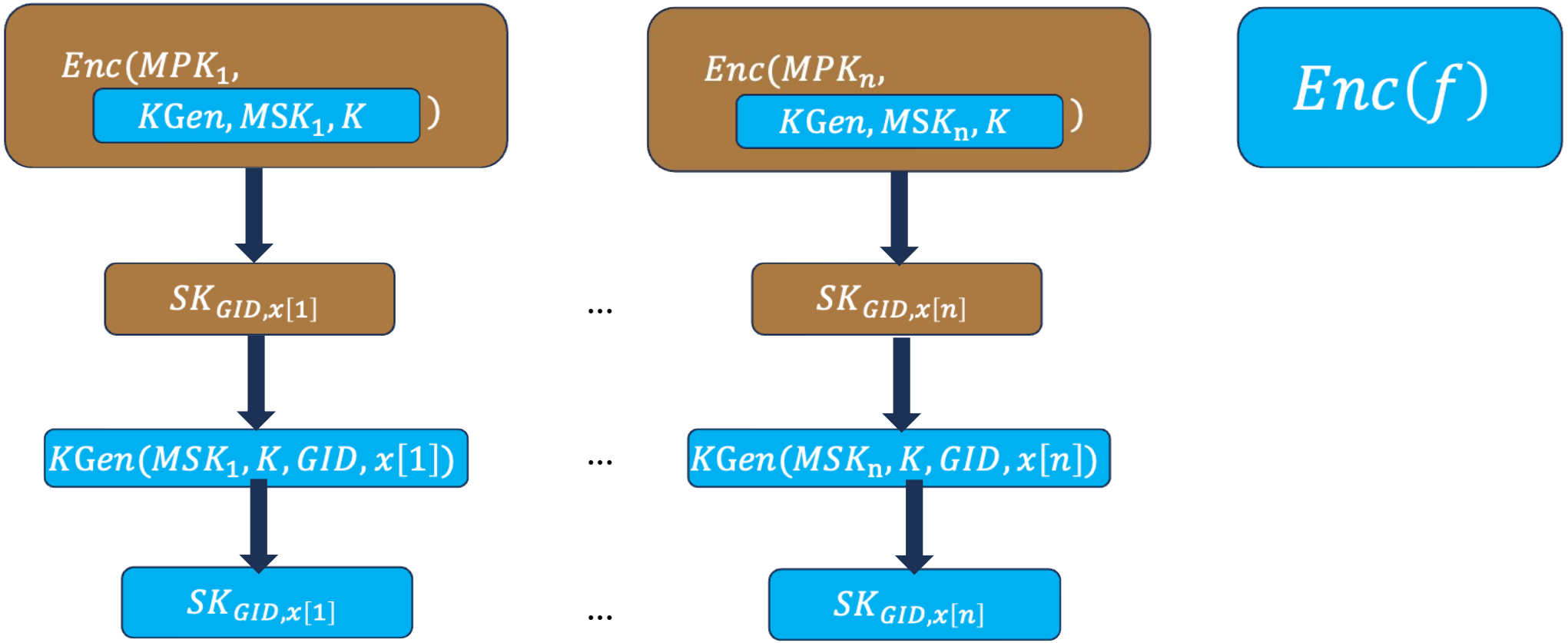


- Output Ciphertext

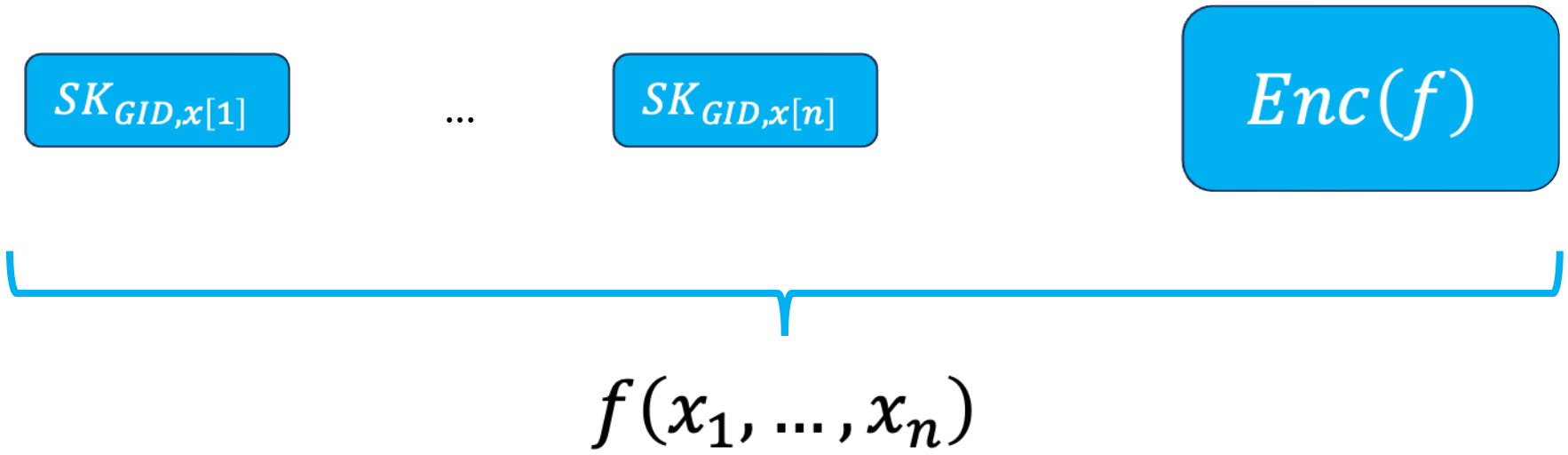


Decryption

- Ciphertext



Decryption



Security

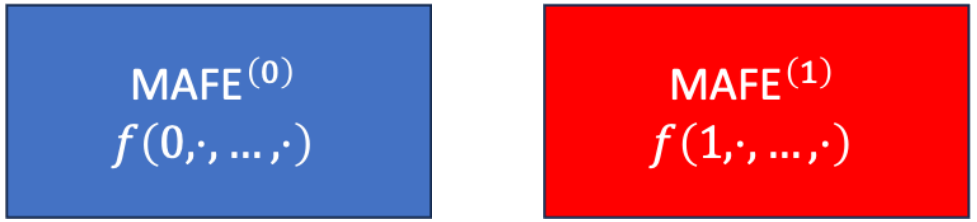
- Like hybrid encryption, two step argument.
- Firstly, argue security of all 1-Auth FE schemes.
  - Now, we only use outputs, $SK_{GID,x}[1]$
- Now, invoke security of $Enc(f)$

2. Authority Corruption Compiler

- Hypothetically, consider Authority #1 will be corrupted
- Functionality wise, such an MAFE scheme reveals

$$f(0, x[2], \ldots, x[n]) \;\&\; f(1, x[2], \ldots, x[n])$$

- To emulate this functionality,



Authority Corruption Compiler

Security: Guess an $i^* \in \binom{n}{k}$ and rely on that MAFE's security

Caveat: Cannot encrypt same f in all MAFE instantiations

Solution: Use Function Secret Sharing with Strong Simulation Guarantees [Benhamouda-Lin20]

❖ For any *k-out-of-n* arbitrary corruptions with corrupted authorities combinedly holding at most $L = \sum_{id} \ell_{id}$ input bits, create

$$\text{MAFE}^{(i,j)} \quad \text{for} \quad i \in \binom{n}{k} \;\&\; j \in \{1, \ldots, 2^L\}$$

Thank You!

MAFE for General Circuits from Polynomially Hard Assumptions

Bounded Collusion

- With adaptive authority corruptions* from LWE or SXDH
- Without authority corruptions from PKE

Collusion Resistant

- With adaptive authority corruptions* from FE + LWE/SXDH
- Without authority corruptions from FE

Additional Results

Generic Compilers & Lower Bounds for authority corruptions

https://eprint.iacr.org/2025/412

Follow Ups

I. MA-ABE for NC^1 from LWE
II. MA-TT from Pairings + LWE
Concurrent: [Datta-Schädlich-Tairi'26]

* For certain parameter ranges